

Architecture Design Report

This document presents the architectural design of the system, including its context, logical structure, runtime behavior, and deployment details.

1. System Requirements

System Name: ELearningManagementSystem

Architecture Style: Microservices Architecture

Functional Requirements

- User registration and authentication
- Manage courses and lessons
- Enroll students in courses
- Stream video lectures
- Track student progress and grades
- Send notifications to students and instructors

Non-Functional Requirements

- Scalability: High
- Availability: High
- Latency: Medium
- Reliability: High
- Consistency: Eventual
- Security: High

2. Formal Architecture Specification (ACME ADL)

The following section provides a formal architectural specification using ACME ADL, capturing components, connectors, and constraints.

```
System ELearningManagementSystem {
  Property view : String = "runtime";
  Property architectureStyle : String = "Microservices Architecture";
  Property source : String = "AI";
  Component UserService = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Handle user registration, authentication, and profile management";
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
  }
  Component CourseService = new Component {
    Port in;
    Port out;
```

```

    Property responsibility : String = "Manage courses, lessons, and course enrollments.";
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
}
Component VideoStreamingService = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Stream video lectures and handle video playback.";
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
}
Component ProgressTrackingService = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Track student progress and grades.";
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
}
Component NotificationService = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Send notifications to students and instructors.";
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
}
Component APIGateway = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Handle incoming requests, route them to the appropriate service";
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
    Property type : String = "L7";
    Property healthChecks : Boolean = true;
}
Component Database = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Store user, course, lesson, and progress data.";
    Property state : String = "stateful";
    Property scaling : String = "vertical";
    Property consistency : String = "eventual";
    Property replication : String = "multi-node";
}
Attachment APIGateway.out to UserService.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = false;
    Property backpressure : String = "bounded";
}
Attachment APIGateway.out to CourseService.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = false;
    Property backpressure : String = "bounded";
}
Attachment APIGateway.out to VideoStreamingService.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = false;
    Property backpressure : String = "bounded";
}

```

```

Attachment APIGateway.out to ProgressTrackingService.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = false;
    Property backpressure : String = "bounded";
}
Attachment APIGateway.out to NotificationService.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = false;
    Property backpressure : String = "bounded";
}
Attachment UserService.out to Database.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = true;
    Property backpressure : String = "bounded";
}
Attachment CourseService.out to Database.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = true;
    Property backpressure : String = "bounded";
}
Attachment VideoStreamingService.out to Database.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = true;
    Property backpressure : String = "bounded";
}
Attachment ProgressTrackingService.out to Database.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = true;
    Property backpressure : String = "bounded";
}
Attachment NotificationService.out to Database.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = true;
    Property backpressure : String = "bounded";
}
Attachment APIGateway.out to Database.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = true;
    Property backpressure : String = "bounded";
}
Attachment UserService.out to ProgressTrackingService.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = false;
    Property backpressure : String = "bounded";
}
Attachment CourseService.out to ProgressTrackingService.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = false;
    Property backpressure : String = "bounded";
}
Attachment VideoStreamingService.out to ProgressTrackingService.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = false;
    Property backpressure : String = "bounded";
}
Attachment NotificationService.out to ProgressTrackingService.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = false;

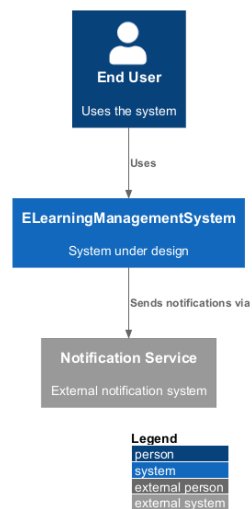
```

```

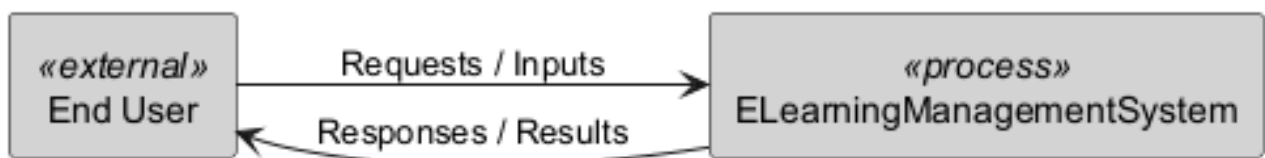
    Property backpressure : String = "bounded";
}
Attachment ProgressTrackingService.out to NotificationService.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = false;
    Property backpressure : String = "bounded";
}
Property resilience_pattern : String = "Retry, CircuitBreaker";
Property delivery : String = "at-least-once";
Property ordering : Boolean = false;
Property authentication : String = "service-to-service";
Property authorization : String = "RBAC";
}

```

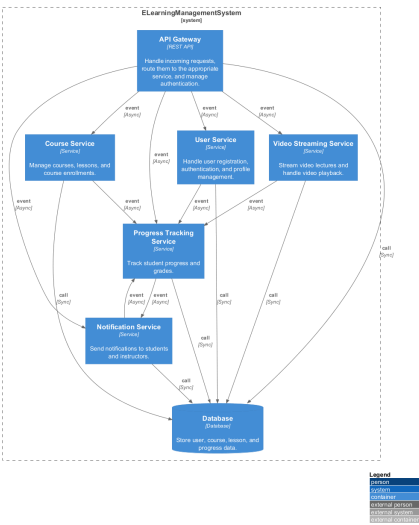
3. Context View



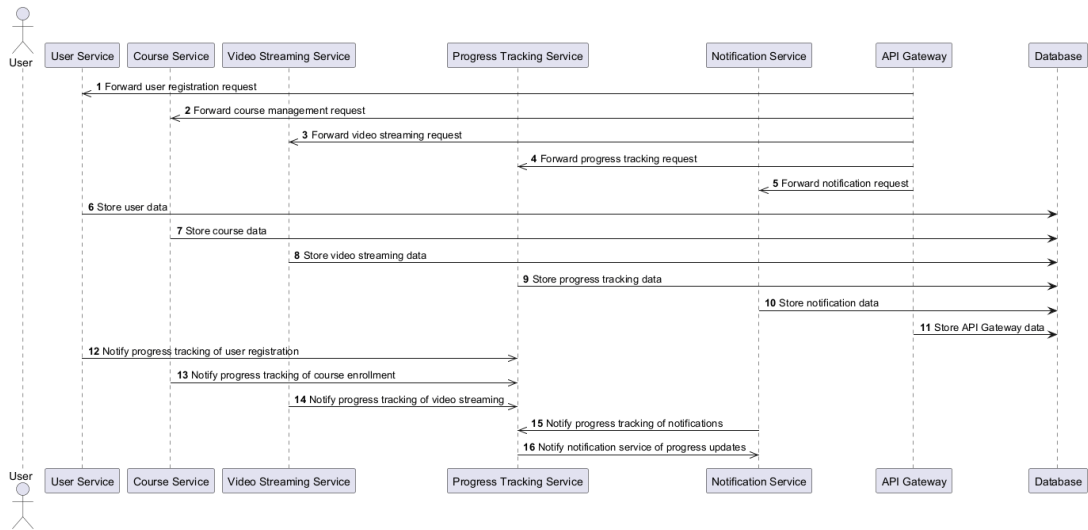
3.1 Data Context View (Level 0 DFD)



4. Logical View (C4 Container Diagram)



5. Process View (Runtime Interaction)



6. Physical View (Deployment Diagram)

